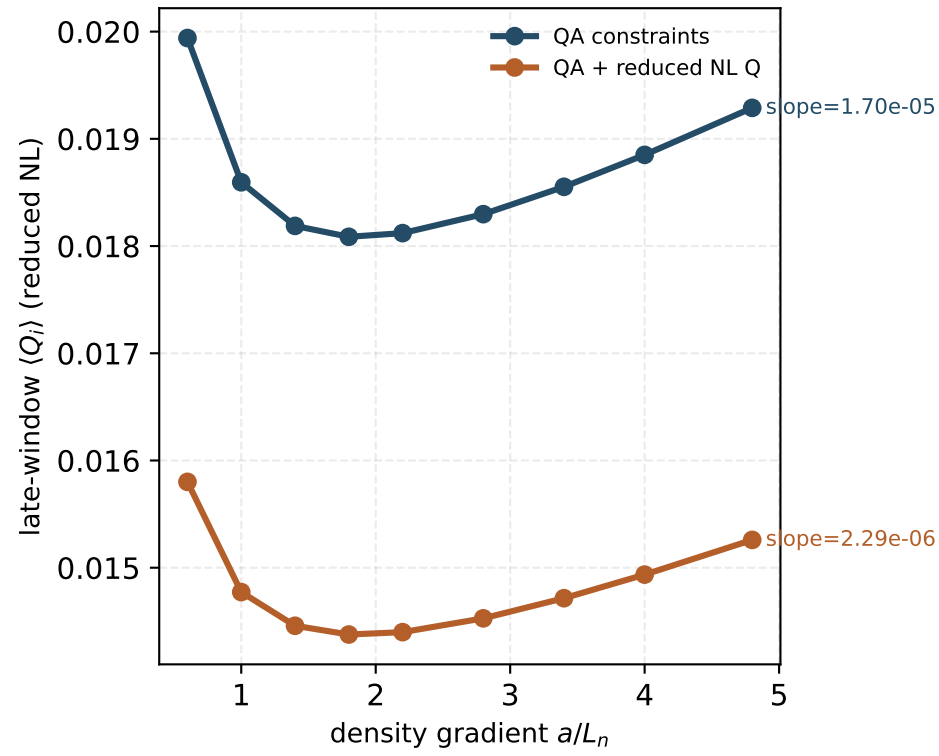
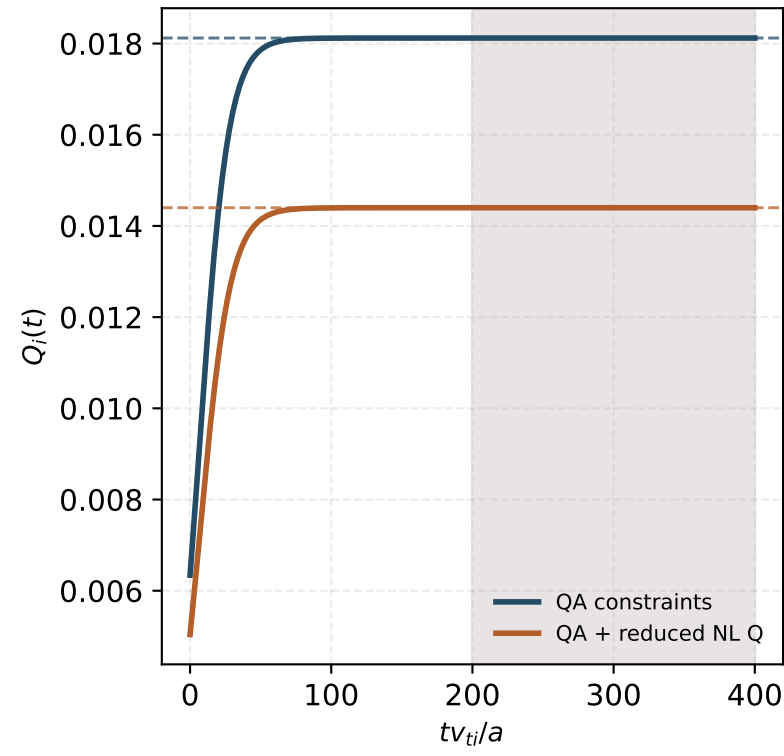


Aspect-6 QA low-turbulence optimization: constraints-only vs transport-aware design

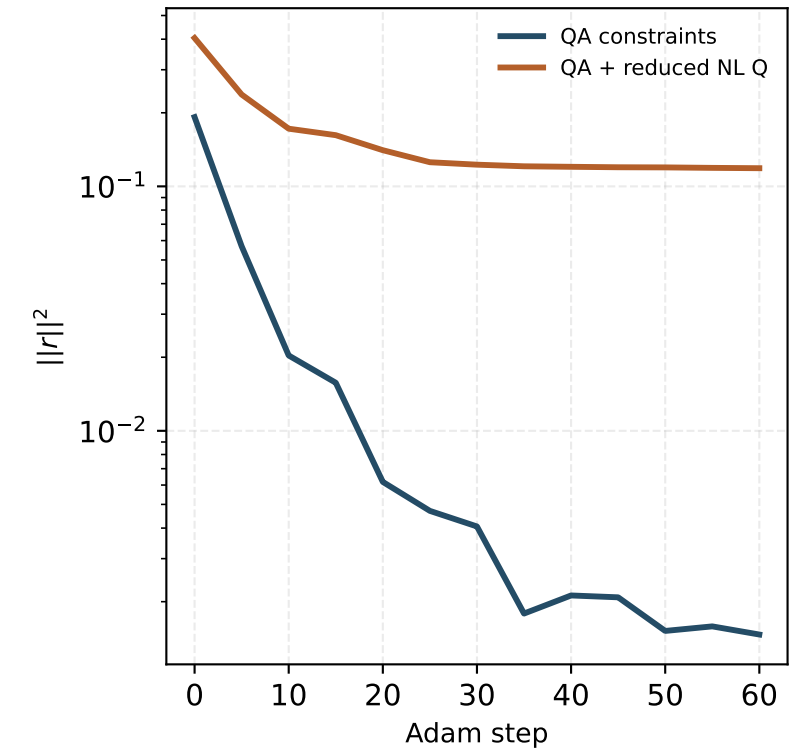
Gradient scan at fixed $a/L_{Ti} = 6$



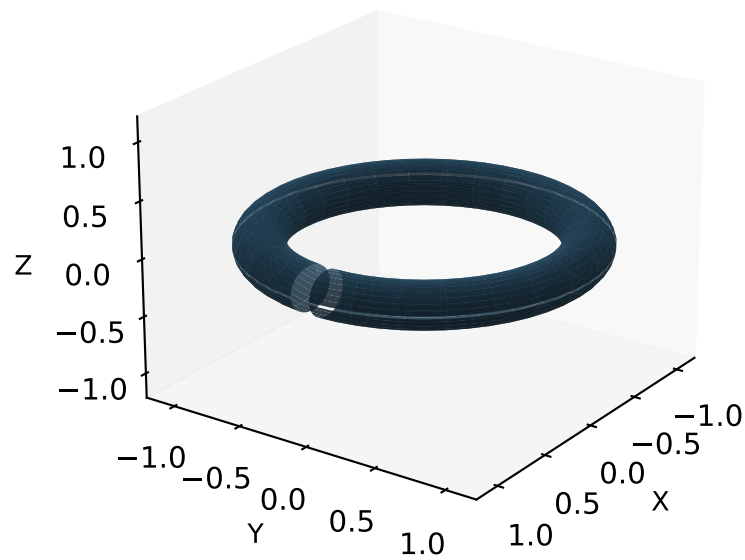
Fixed-gradient trace: $a/L_n = 2.2$, $a/L_{Ti} = 6.0$



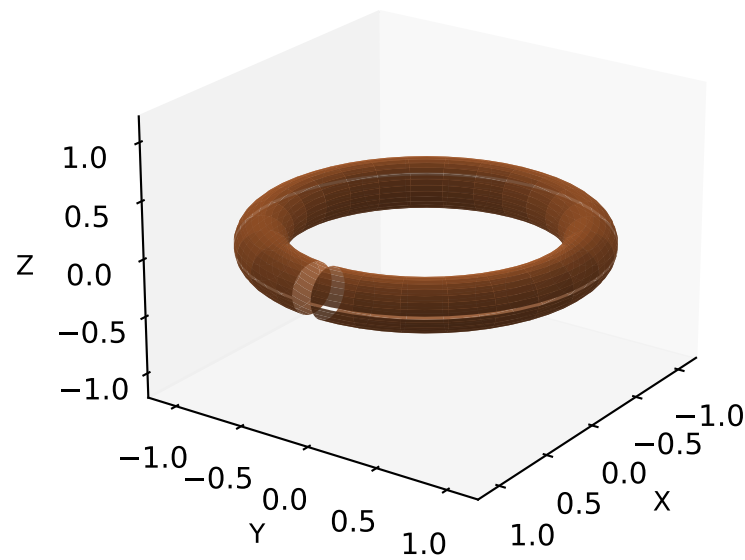
Constrained objectives



Reduced LCFS: QA constraints



Reduced LCFS: QA + reduced NL Q



Gate summary

Aspect target: 6.0

Minimum iota: 0.41

Operating iota floor: 0.70

AD/FD gates passed: True

Constraint gates passed: True

Long-window gates passed: True

Fixed-gradient Q reduction: 20.5%

Trace tmax: 400

Final values

QA constraints

A=5.986, iota=0.739

QA residual=1.99e-03

$\langle Q_i \rangle = 1.812e-02$

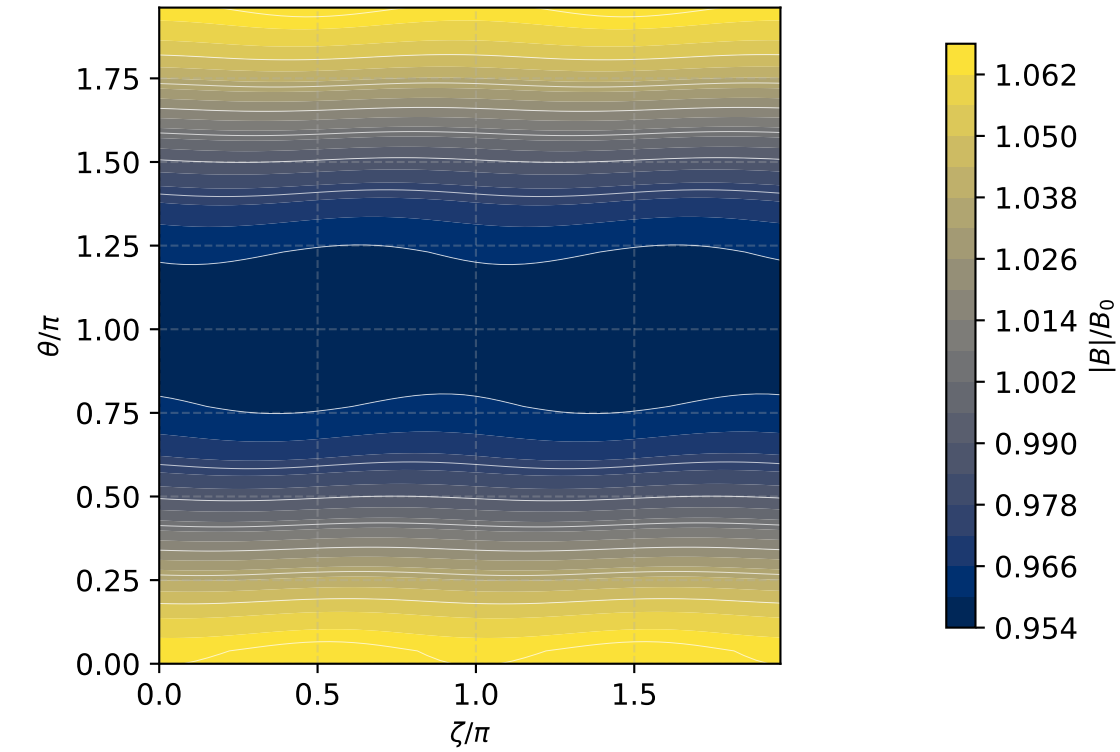
QA + reduced NL Q

A=5.976, iota=0.756

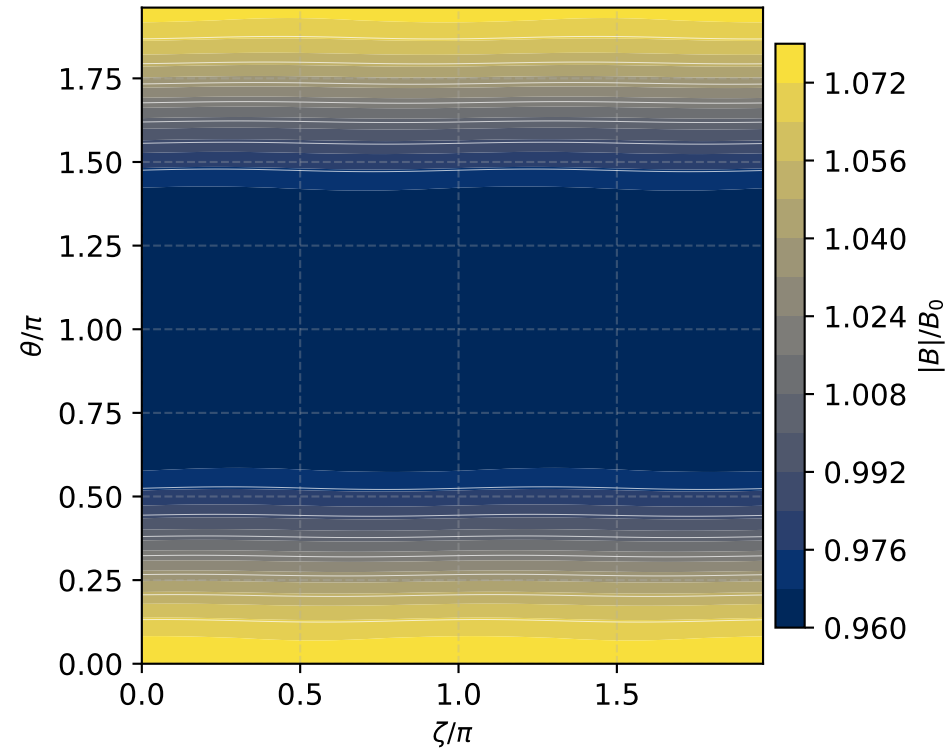
QA residual=1.01e-03

$\langle Q_i \rangle = 1.440e-02$

LCFS $|B|$: QA constraints



LCFS $|B|$: QA + reduced NL Q



Scope and method

- Reduced max-mode-1 QA controls
- Aspect-ratio and two iota-floor constraints
- Long differentiable RK2 nonlinear ITG envelope
- Scalar/residual/observable AD-FD gates
- Production nonlinear claims still need long-window GK audits

Equations used in this panel

$J = \|r_A, r_i, r_{QA}, r_{reg}, r_Q\|^2$

$dE/dt = 2\gamma E - \alpha E^2$

$Q_i = W_i E$